

Formal Verification: 6CCS3VER

Exercise Sheet 6

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 6.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

```
lemma
  assumes T: "T b"
    and A: "A a  $\wedge$  A b  $\implies$  a = b"
    and TA: " $\bigwedge x. T x \implies A x$ "
    and Aab: "A a"
  shows "T a"
```